

From Cloud to Edge: Powering Azure Services Anywhere



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Global Principal Architect

Microsoft Azure MVP since 2018

- Azure Hybrid & Migration, Azure Networking

Focus Areas

- Cloud Adoption, Strategy and Architecture
- Cloud Platform engineering and governance
- DevOps & Developer Enablement
- Sovereign Hybrid Cloud



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Microsoft Sovereign Cloud

Most comprehensive set of sovereignty solutions across
AI, productivity, security and cloud platform

Sovereign Public Cloud

Data stays in Europe, under European law

Operations and access controlled by
European personnel

AI model choice and AI processing that
stays in Europe

Sovereign Private Cloud

Continuous operations across hybrid or
disconnected environments

Integrated cloud and productivity with
Microsoft 365 Local

AI capabilities and advanced GPU support
with Azure Local

National Partner Clouds

Access in-country expertise and
nationally owned organizations

Germany: Delos designed to
meet BSI Cloud requirements

France: Bleu designed to meet
SecNumCloud requirements

Consistent infrastructure management and development platform

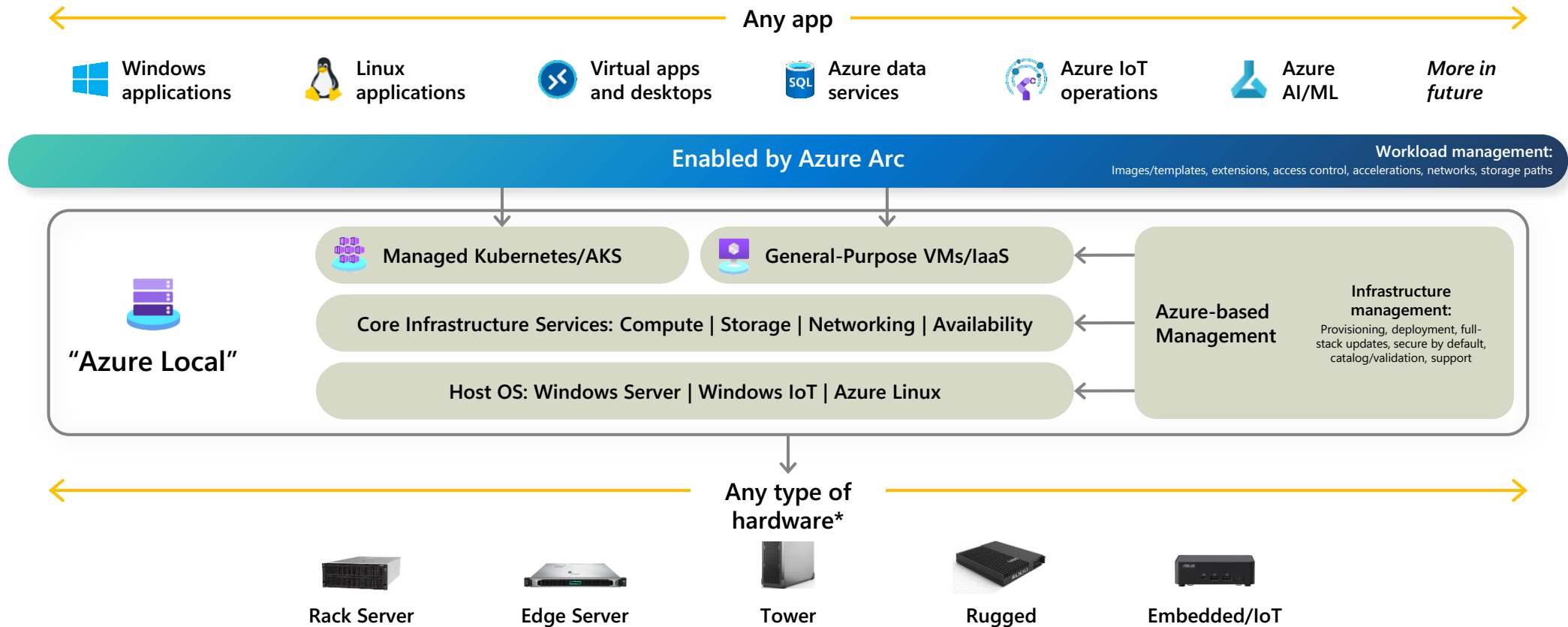
<https://blogs.microsoft.com/on-the-issues/2025/04/30/european-digital-commitments/>

<https://blogs.microsoft.com/blog/2025/06/16/announcing-comprehensive-sovereign-solutions-empowering-european-organizations/>

<https://azure.microsoft.com/en-us/blog/microsoft-strengthens-sovereign-cloud-capabilities-with-new-services/>

Azure Local: One flexible offering for all target use cases

Unified distributed infrastructure service spanning all hardware/scale points



*Must meet minimum requirements per operating system and solution-level pass validation

Any Size – Any Where



Consistent software platform, Portal, and APIs



Low-spec, low-cost edge servers

Simpler, smaller hardware for light computing requirements like SE30, SE50, SE350v2

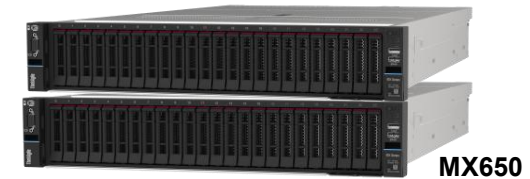
NEW PREVIEW



Connected servers

Choose from over 20 hyperconverged server platforms from Lenovo

✓ GA



Disconnected operations

Meet strict data residency regulations with a permanently disconnected option.

✓ GA

Low-spec, low-cost options for edge use cases

Azure Stack HCI Requirements at launch	Azure Local Requirements at launch
Windows Server certified	Windows Server certified
Min. 2+ machines	1+ machine
Min. 4+ disks per machine	1+ SSD per machine
Min. 10 Gbps w/ RDMA	1 Gbps/2.5 Gbps Ethernet
Active Directory required	Doesn't require AD ¹



ThinkEdge SE30
Fan-less server

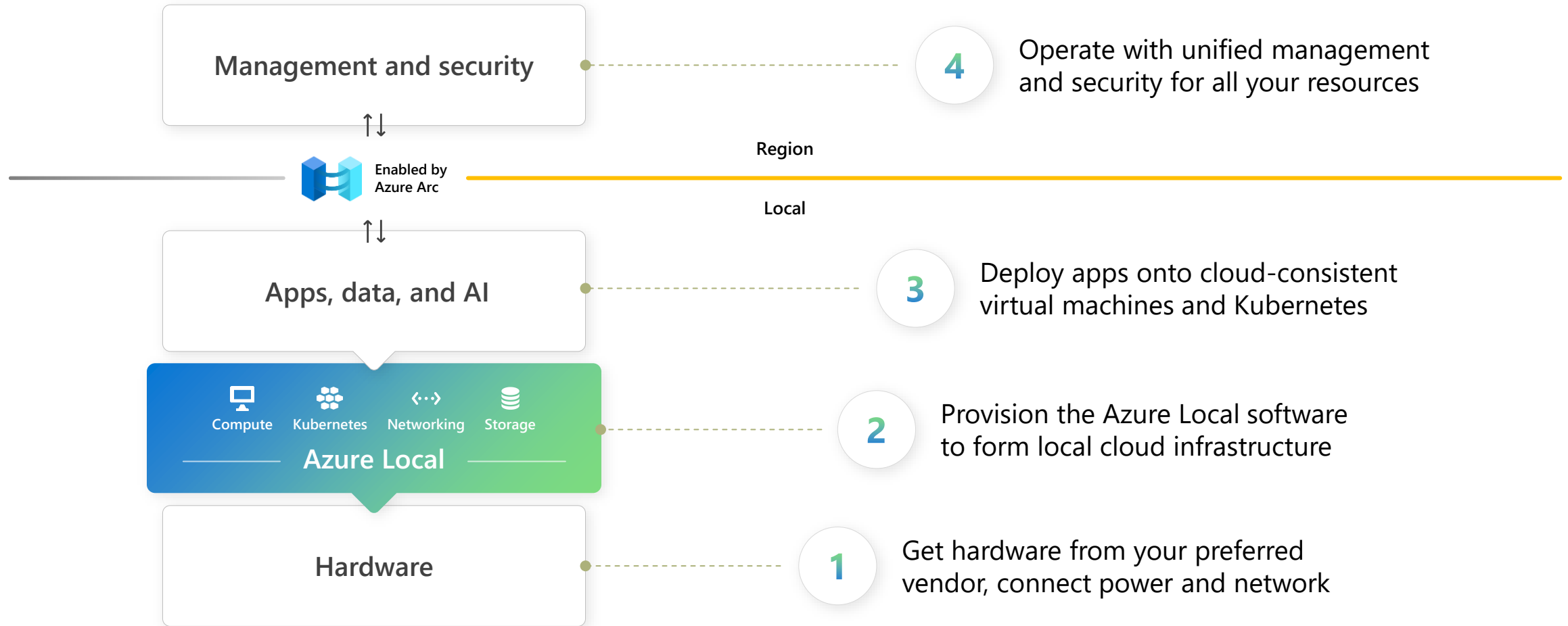


ThinkEdge SE350v2
Half-width, half-depth 1U



ThinkEdge SE100
GPU attachable

How Azure Local works (connected)



Bring Azure app, data, and AI services anywhere

Management services



Portal



Copilot



Graph



Identity



Defender



Monitor



Updates



Policy



Support



Billing



Popular



Windows
and Linux



Azure Virtual
Desktop



Azure IoT
Operations



Container
Apps



Logic Apps

Data services



Arc-enabled
SQL Server



Managed
instance

AI services



Video
Indexer



Foundry Local
(preview)



Agentic
Retrieval
(preview)



Machine
Learning

Additional



IoT
Operations



Event Grid



M365



GitHub
(preview)

Foundational services



Virtual
machines



Kubernetes
services



Logical
networks



Storage
paths

Azure Kubernetes Service built-in and included



Managed Kubernetes with AKS-consistent portal and APIs to simplify operations



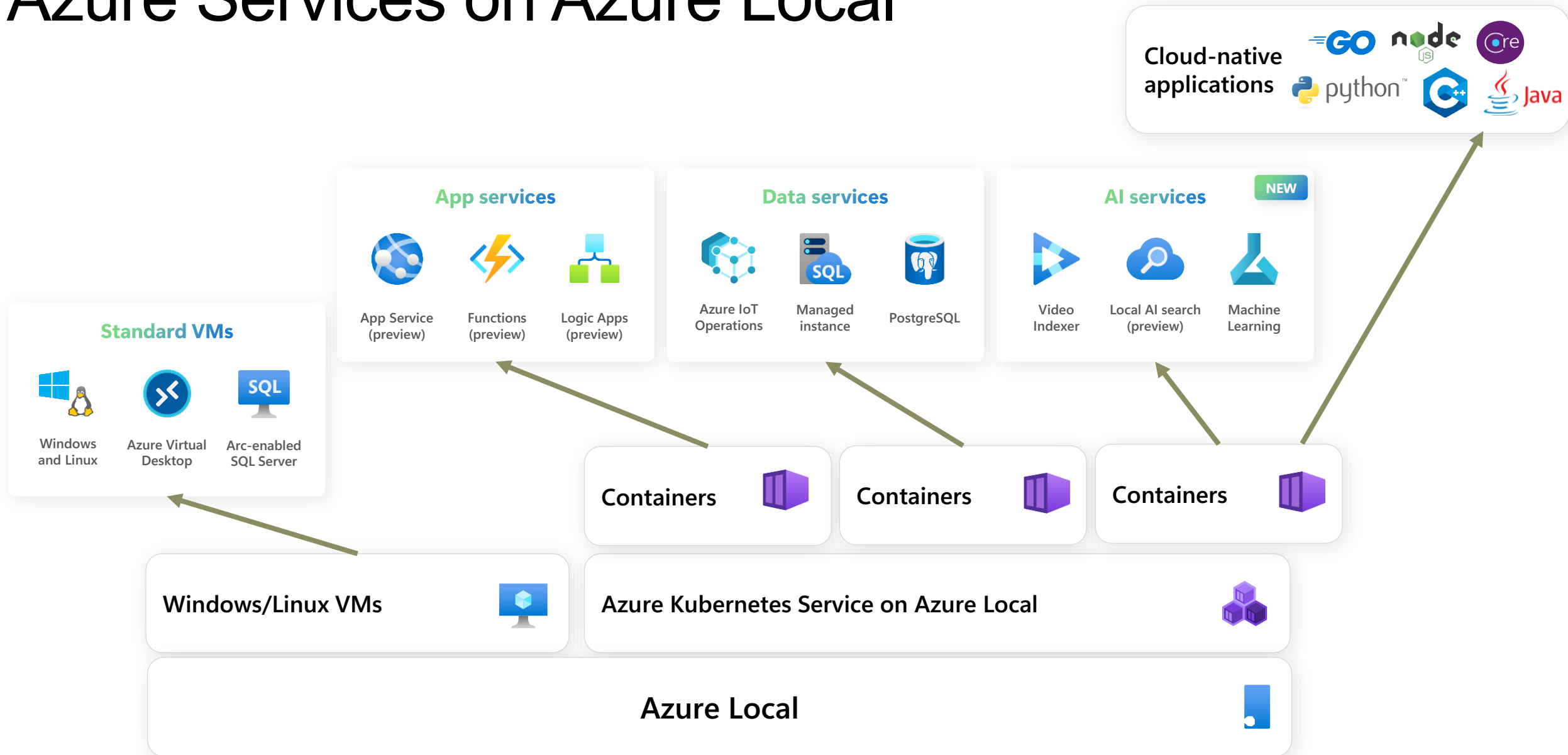
Microsoft-provided storage (CSI) driver and images for Linux and Windows



Infra-as-code and continuous delivery with Terraform and GitOps support

The screenshot displays the Microsoft Azure portal interface. The top navigation bar shows 'Microsoft Azure' with a search bar and a 'Copilot' button. The breadcrumb trail indicates the path: 'Home > Azure Arc > Azure Local > Contoso-01 >'. The main heading is 'Create a Kubernetes cluster with Azure Arc'. Below this, there are tabs for 'Basics', 'Node pools', 'Access', and 'Networking'. The 'Basics' tab is active, showing 'Project details' where a subscription and resource group are selected. The 'Cluster details' section includes fields for 'Kubernetes cluster name' (My-AKS-Cluster), 'Custom location' (Contoso-01), and 'Kubernetes version' (1.28.9). The 'Primary node pool' section shows 'Primary node pool contains Kubernetes Linux nodes that', 'Node size' (Standard_A), and 'Node count' (1). A 'Review + create' button is at the bottom. An overlay window shows the 'My-AKS-Cluster' resource page, which includes a left-hand navigation menu with options like 'Security (preview)', 'Namespaces', 'Workloads', 'Services and ingress', 'Storage', 'Configuration', 'Settings', 'Node pools', 'Extensions', 'Open Service Mesh', 'GitOps', 'Networking', 'Policies', 'Monitoring', 'Insights', 'Alerts', 'Metrics', 'Logs', 'Diagnostic settings', and 'Workbooks'. The main content area of the overlay shows 'Essentials' with details like 'Subscription: Contoso', 'Subscription ID: ebaf548b-cb71-4663-9cda-a42bfa0c7bc9', 'Resource group: Contoso-01-RG', 'Status: Connecting', and 'Location: Contoso-01 (East US)'. It also shows 'Tags' and 'Properties' for the 'Kubernetes service - Azure Arc' resource, including 'Agent version: 1.21.10', 'Managed identity certificate: January 5, 2025 at 10:21 AM PST', 'Kubernetes version: 1.29.4', 'Total node count: 5', and 'Total core count: 20'.

Azure Services on Azure Local



Sovereign Local AI – Done easy

Sovereign Private Cloud AI stack

AI Services



BYO RAG

Bring your own RAG application and logic



Edge RAG

Search and analyze local knowledge, including SharePoint data



AI Video Indexer

AI video and audio insights from live* or recorded sources

Foundry Model Catalog



OpenAI OSS



Call



Mistral



DeepSeek



Microsoft

Task-specific models

OCR, speech, vision, translation, and more

BYO models

Foundry Local Model Server

(inference and runtime deployment*)

AI Infrastructure

NVIDIA GPUs

Leading OEM Partners

Dell, HPE, Lenovo, and more

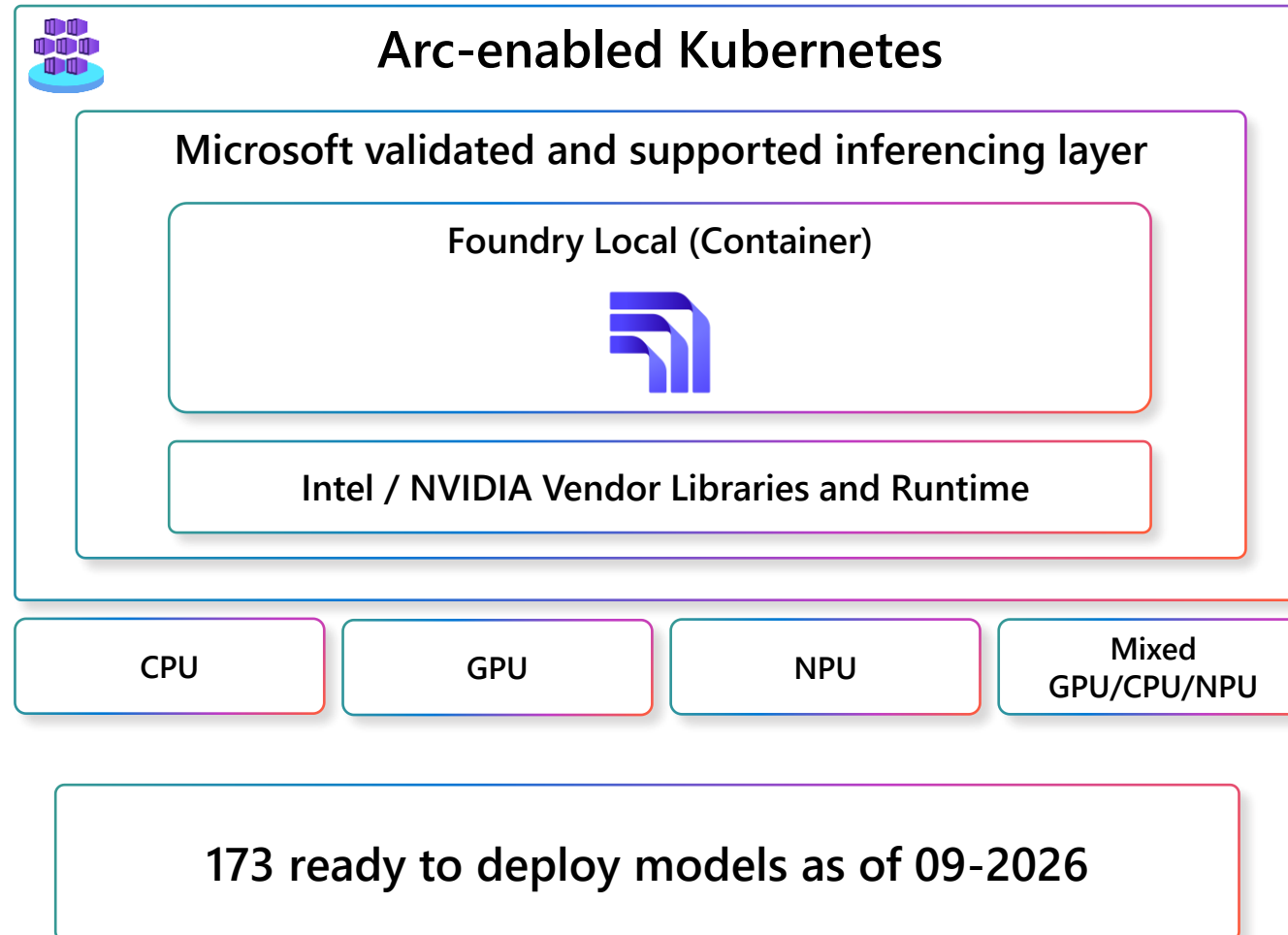
Small to large footprints

Single node to multi-rack deployments

← Connected and fully disconnected operations with Azure Local →

Designed to run 1000+ models such as GPT OSS, DeepSeek-V3, Mistral NeMo, Llama 4 Maverick, RTX Pro 6000 BSE enables organizations to accelerate their AI initiatives directly within a sovereign private cloud environment.

Foundry Local on Kubernetes



Edge RAG

Preview

Intelligent Knowledge Handling



- LazyGraphRAG for high-quality retrieval
- OCR-based parsing of documents, tables, and images, with SharePoint integration.



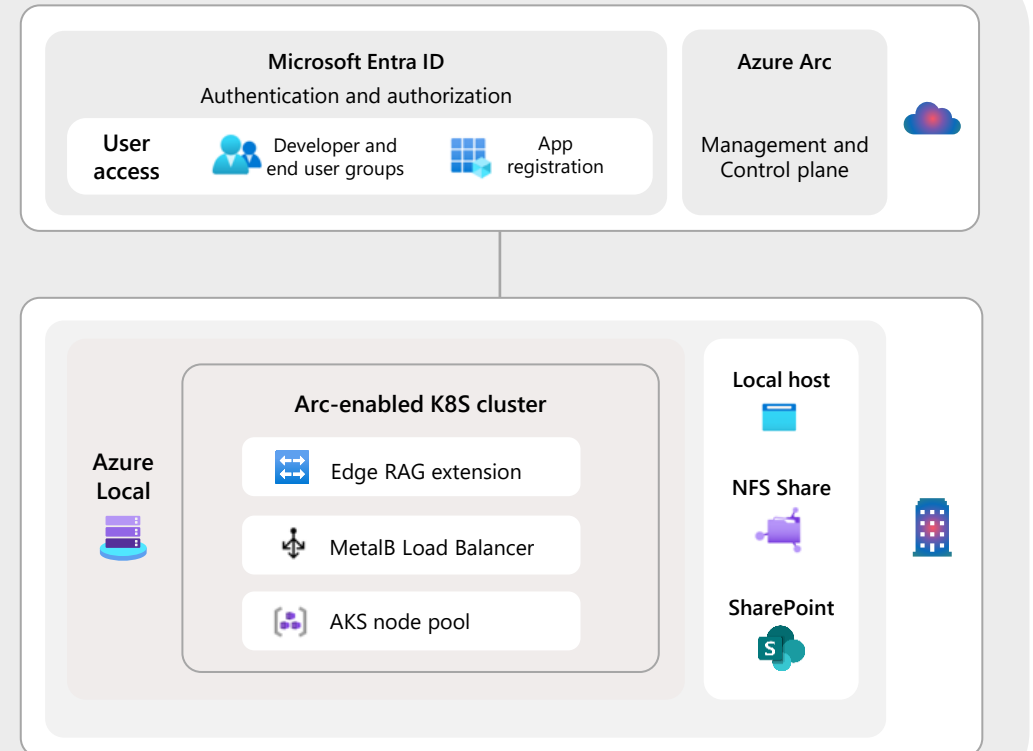
- 100x faster ingestion of live-streamed images
- 5x faster query performance for Hybrid Search



- Multimodal search, image retrieval & image-rich outputs with chat UI upgrades
- Connected and fully disconnected scenarios



- Agentic workflow with OCR & LLM integration focusing on CEQA (California Environment Quality Act) applications



Vision AI: Azure AI Video Indexer

Preview

Enable Real-Time Video Intelligence



Create custom AI models with natural language and apply object detection in restricted areas.



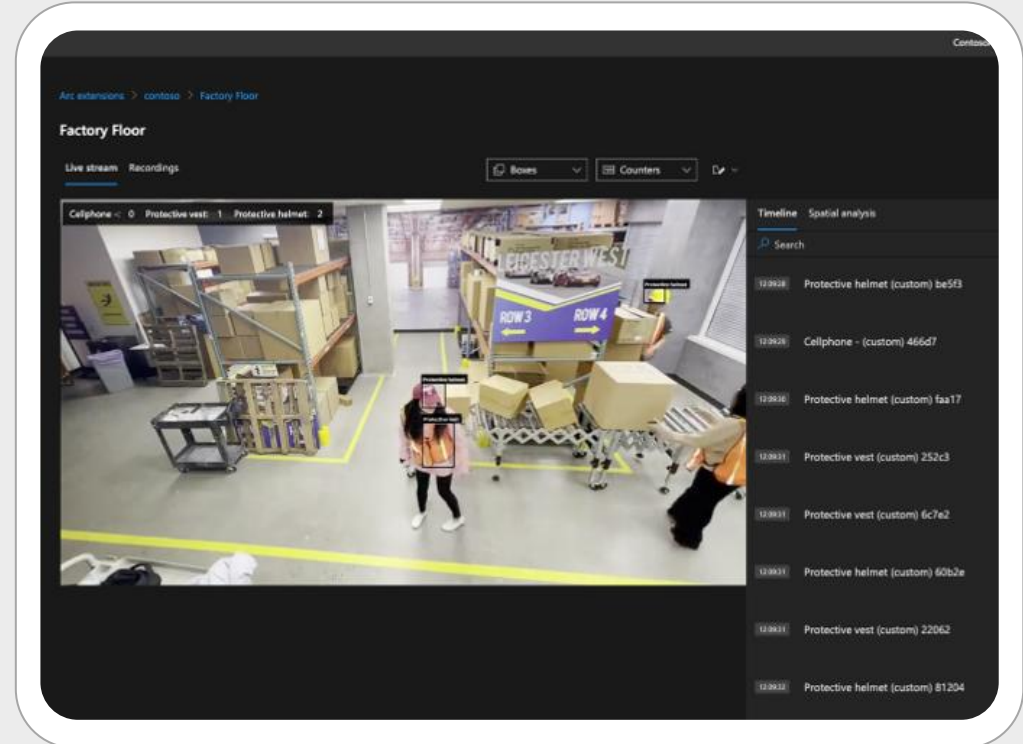
AI video assistant chat: Ask questions in natural language and get real-time answers



Video agents for retail, service, security, and sales to monitor safety, experience, and performance



Capture key events and insights using natural language to generate summaries

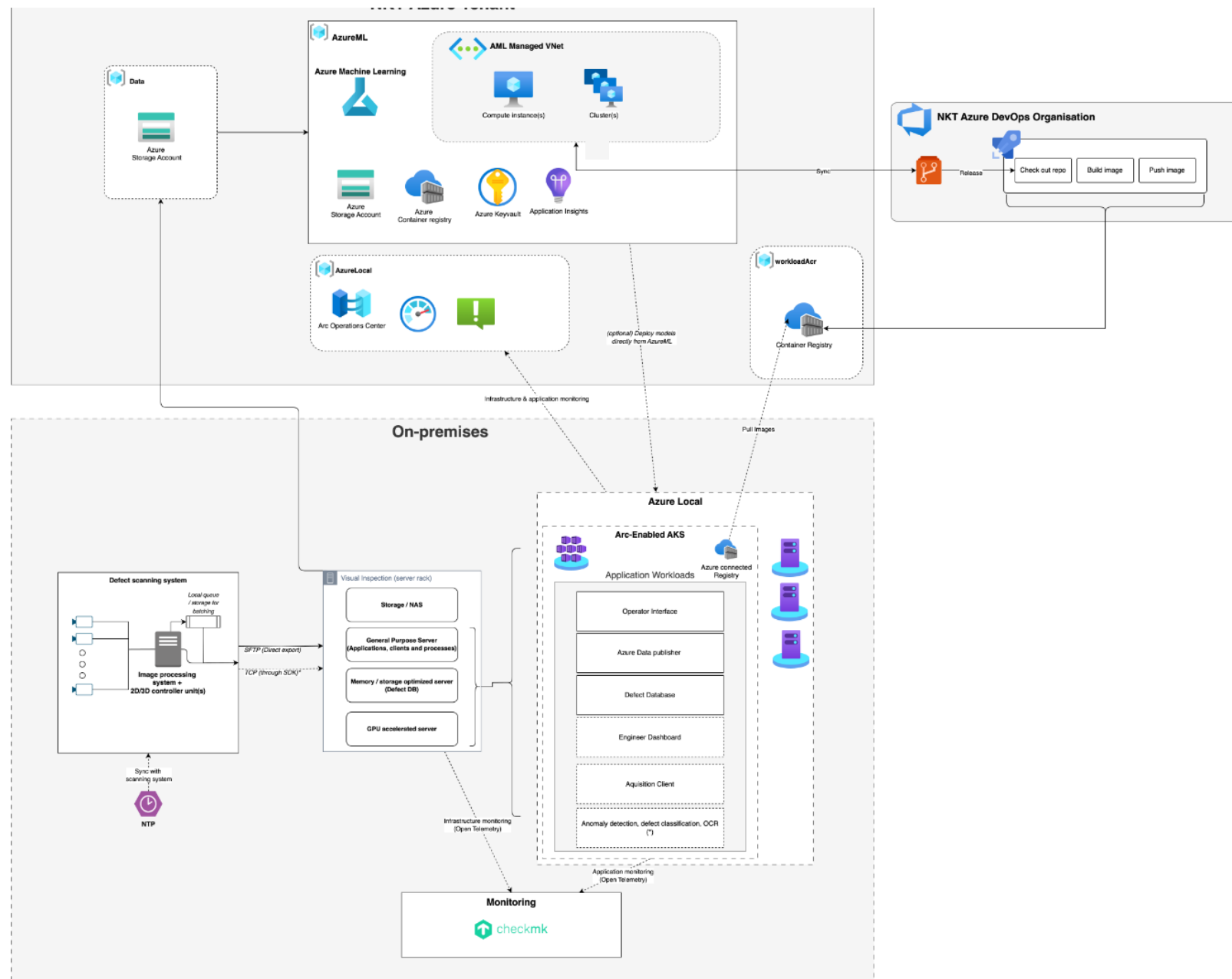


Hybrid Image ML

An **hybrid architecture** that effectively combines and connects on-premises infrastructure and scanner hardware with Azure Cloud.

The solution consist of three main components or layers:

- I. Defect scanning system
- II. On-premises architecture – Dataprocessing workloads, applications and hybrid hosting
- III. Azure cloud architecture - AI development platform, long term data storage and centralized governance and monitoring



DEMO

Disconnected operations

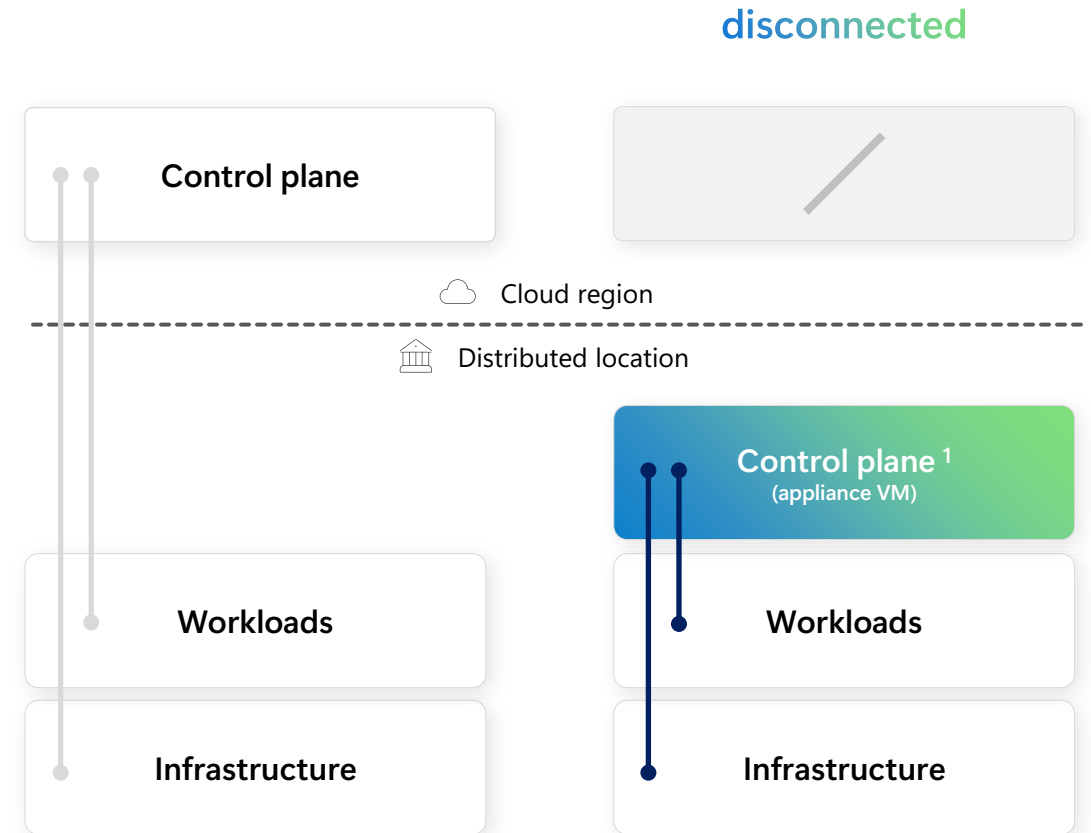
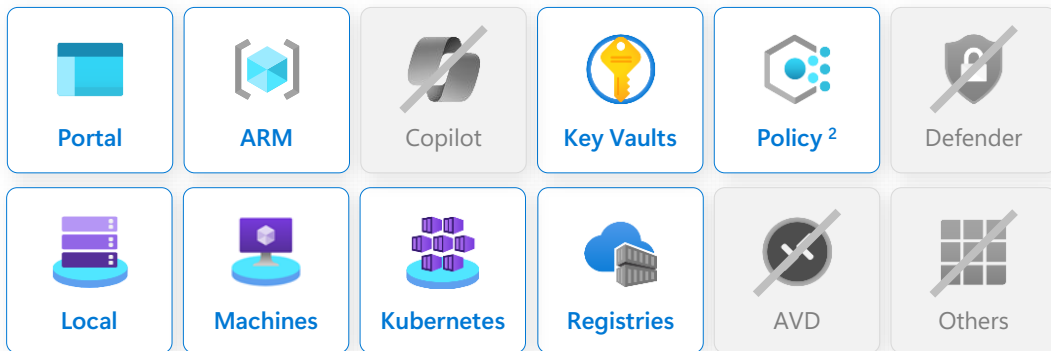


Satisfy regulatory requirements by operating permanently disconnected from the cloud



Host backend Azure resource manager, portal, and services in local appliance VM

Subset of services available:



1 : Available only to customers who prequalify based on industry, use case, and other considerations 2 : Partial functionality

Microsoft Sovereign Cloud

Public Cloud

- Cloud Adoption Framework
- Well-Architected Governance, Security & Landing Zones
- Defender for Cloud
- Customer Lockbox
- EU Data Boundary
- Data Guardian

Sovereign Public Cloud

- Sovereign Controls
- Regulated Environment Management
- Customer-managed Keys
- Confidential Compute

Sovereign Private Cloud

- Azure Local Connected
 - Azure Kubernetes Service + some Azure services
- Microsoft 365 Local
- (GitHub Enterprise)

Disconnected

- Azure Local Disconnected
 - Azure Kubernetes Service + some Azure services
- Microsoft 365 Local
- (GitHub Enterprise)

Questions?

twoday